

UNI HACK SUBMISSION

AI-Powered Product Intelligence for Industrial Commerce

Turning a 6-column part row into a 252-column, source-verified, commerce-ready record — with provenance and confidence on every field.



Data Enrichment



Source Validation



Explainable Outputs

Verified on an industrial abrasives dataset · Node.js + SQLite pipeline · React explainability UI

THE PROBLEM

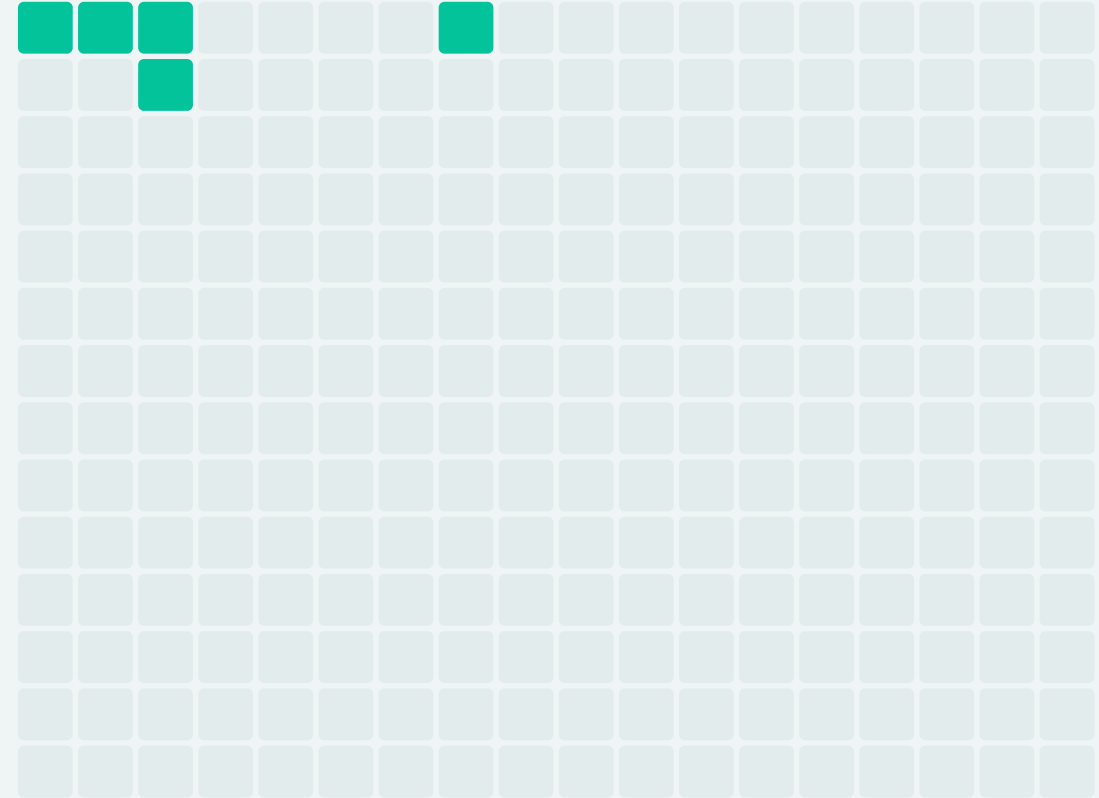
Raw part data is sparse, inconsistent, and unverifiable

- Input is just 6 columns per part — and brand fields are often placeholders like "-- Unbranded --" or "-- No Unilog Brand --".
- Commerce systems need 252 structured fields: classification, attributes with units, six description variants, identifiers, media.
- Naive AI simply "fills the gaps" — no source, no confidence, no way to audit whether a value is real or invented.

6 → 252

columns to populate, from source-verified evidence

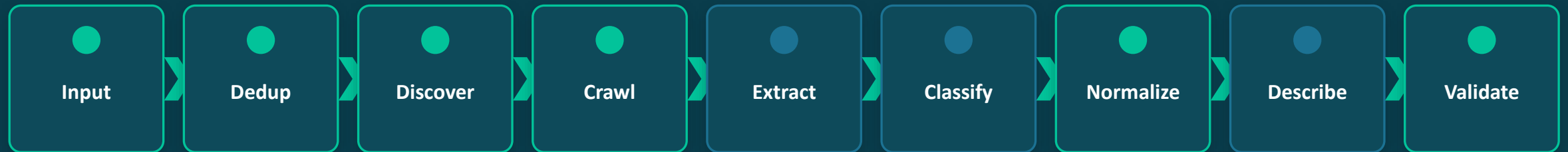
THE 252-FIELD TARGET — EMPTY AT INPUT



5 known · 247 missing

A source-verified enrichment pipeline — every field traceable

Give it whatever the catalog has. Ten cooperating phases discover official sources, read them, extract with evidence, then normalize and validate deterministically — emitting confidence and a needs-review signal on every value.



Export → 252-column Unilog CSV

● AI stage

● Deterministic (no AI)

Manufacturer-first

Marketplaces blocked; official domains resolved by score

Evidence on every field

Source URL, snippet, role, confidence, found-vs-inferred

Rules, not guesses

Normalization & validation are fully deterministic

Find and read the right sources

Three deterministic stages — no LLM — prepare clean, official evidence before any AI runs.

Deduplicate

- ✓ Keyed on a normalized manufacturer part number.
- ✓ Duplicates link to a canonical product and inherit its enrichment — no re-runs.

LIVE · 11 of 29 reused live

Discover source

- ✓ Manufacturer-first search; each domain scored (exact +0.8 ... accept ≥ 0.6).
- ✓ 30-domain blocklist removes Amazon, Grainger, McMaster, DigiKey, Alibaba...

LIVE · 8 resolved · 6 success

Crawl & read

- ✓ Cheerio → Playwright escalation for empty SPAs or sparse pages.
- ✓ Parses PDFs (pdf-parse) & DOCX (mammoth); 1 request / domain, SSRF-guarded.

LIVE · adaptive engine

Every extracted value carries its evidence

AI STAGE

model: gemini-3.5-flash-lite

- An LLM reads sanitized source content and returns attributes against a JSON schema (temp 0.1).
- Each value records source URL, source name, role, evidence snippet, confidence, and found-vs-inferred.
- Disagreements become explicit conflicts with a written resolution — it never silently picks a winner.

REAL ATTRIBUTE · product 49 (3M Cubitron II disc)

Unit of Measure = Case

source_name: **Jam Industrial Supply LLC**
 source_role: **part_manuf** confidence: **1.0**
 snippet: "Unit of Measure Case"

CONFLICT DETECTED & RESOLVED

Attachment Type

PSA (jamindustrialsupply.com) vs **Hook & Loop** (3m.com)

Resolution: "Stikit implies PSA for this variant line, whereas 3M lists general platform options."

resolved · conf 0.8

Structure it — then prove it's commerce-ready



■ AI ■ deterministic

WHAT EACH STEP PRODUCES

Classify

Nearest taxonomy leaf via embeddings + LLM confirm — 6 nodes / 41 attributes.

Normalize

Deterministic rules with a per-value audit trail (e.g. 0.5 → 1/2).

Describe

6 description variants; the invoice line is ALL-CAPS and ≤ 40 chars.

THE COMMERCE-READY GATE

commerce_ready = $\text{conf} \geq 0.70 \cdot \text{no high-severity} \cdot \text{fieldCount} > 0$

24 / 29

products commerce-ready

Confidence clusters at 0.85 · 0.70 · 0.60.

Minor 'error' flags are non-blocking — those products are still commerce-ready.

Built to the brief: enrich, validate, explain

Data Enrichment

- ✓ Discovery + crawl + LLM extraction fills the 6→252 gap
- ✓ Reads official pages, PDFs & DOCX — dedup reuses identical parts
- ✓ Constrained output: placeholders→empty, approved labels only

extractor · orchestration · engine

Validation

- ✓ Deterministic conflict detection with written resolutions
- ✓ Char-limit / LOV / UOM compliance enforced
- ✓ Confidence score + commerce-ready & needs-review gate

validator/agent.mjs

Explainable Outputs

- ✓ Every field traceable to a source snippet + URL
- ✓ Found-vs-inferred flag on each value
- ✓ Explainability drawer + carried into the 252-col export

provenance schema · ExplainabilityDrawer

Concurrent by design, rate-limited for reliability

IMPLEMENTED • VERIFIED

- Product-level worker pool — `PRODUCT_CONCURRENCY` = 10.
- Per-model sliding-60s limiters with token reservation → reconcile.
- Per-domain crawl limiter = 1 concurrent request.
- Automatic fallback model per agent; stale-product recovery on startup.

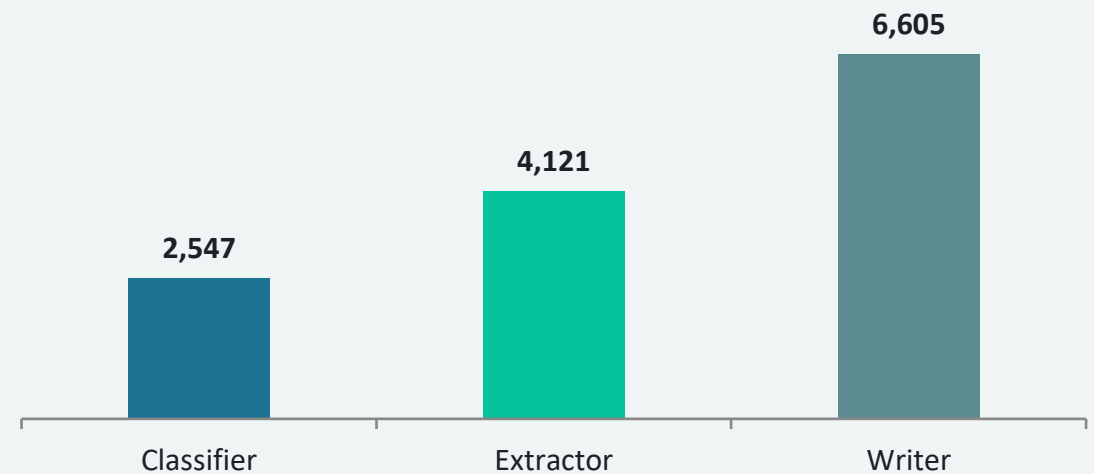
Gemini 12 RPM · 250K TPM · 500 RPD

Qwen 25 RPM · 7K TPM · 1K RPD

Gemma 25 RPM · 14K TPM · 14.4K RPD

Real evidence today: 53 logged LLM calls · 127,647 tokens

AVG LATENCY BY AGENT (ms) · n = 53 calls



Measured over 53 real LLM calls. The writer is the slowest stage — it emits six description variants per product; the classifier is lightest.

What we would tell a technical reviewer

KNOWN LIMITATIONS

- Value-partial: all 252 columns are emitted, but only ~30–40 are typically populated.
 - Finite blocklist — some distributors still surface (e.g. product 49 used a reseller).
 - No benchmark run yet — accuracy against a 200-record ground truth is not yet measured.
- The crawler honours meta-robots directives, not robots.txt.
 - Frontend gaps: upload batch-progress is broken, no charts, and no authentication.
 - Model identifier strings are non-standard — treat them as configured IDs.

Thank you

Data Enrichment

Source Validation

Explainable Outputs